CSC429: Lab #1

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1. Open-Source Data Resources

Resource 1: MNIST

Link: https://www.kaggle.com/datasets/hojjatk/mnist-dataset
Description: The MNIST dataset contains 70,000 handwriting digit samples. Frequently used to evaluate Federated Learning models for image classification.

Resource 2: CIFAR-10

Link: https://www.cs.toronto.edu/ kriz/cifar.html
Description: The CIFAR-10 dataset contains 60,000 image samples with 10 classes. Often used to evaluate Convolutional Neural Networks.

2. Open-Source ML Algorithms

Classification: Convolutional Neural Network (CNN)

Link: Keras Sequential Model
Description: A Convolutional Neural Network (CNN) is a deep learning algorithm that takes input features and passes them through convolutional and pooling operations to detect patterns. Very commonly used for image classification.

Regression: Linear Regression

Link: Scikit-Learn Linear Regression
Description: Linear regression is a foundational ML algorithm used to model outcomes based on one or more independent variables using basically a line. Linear regression algorithms are extremely inexpensive and often able to be trained locally for Federated Learning applications.

Clustering: K Means Clustering

Link: Scikit-Learn K-Means
Description: BriK-Means is an unsupervised ML algorithm that partitions unlabelled data into clusters based on feature similarity. Scikit-Learn has an open-source library that is very handy for clustering tasks.

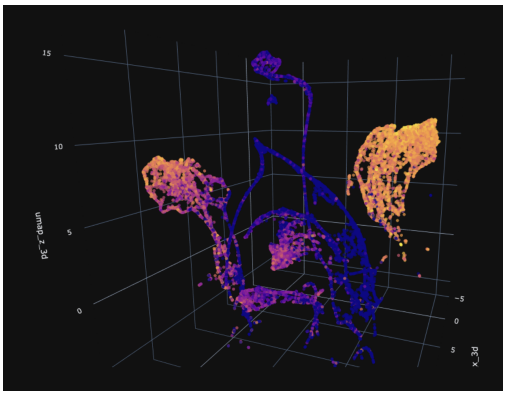

Figure 1: K Means Clustering Example 3d representation

3. Security and Privacy Applications

Application 1: Federated learning in IoT devices.

Federated Learning (FL) allows IoT devices to collaboratively train machine learning models using local data, such as vehicle sensor data or user network traffic, without sending raw data to a central server. ML is very useful here because it enables distributed networks to learn from vast amounts of real-world data while strictly preserving user privacy, as only the computed model updates are shared rather than the sensitive data itself.

Application 2: Privacy-preserving intrusion detection in IoT.

For industrial IoT environments, it is important to have robust Intrusion Detection Systems (IDS). However, oftentimes industrial network traffic is sensitive, and transferring it to a central server risks exposing proprietary operational data. Machine Learning is useful here because models, such as a 1D CNN, can be deployed to monitor and classify malicious traffic (like a DoS) locally. This is where FL becomes particularly valuable. Instead of sending raw industrial network traffic to a central server, each device trains the IDS model locally and shares model updates with the aggregator. However, as we have seen in our papers, even these model updates can leak sensitive information, say, through membership inference attacks. That is why in many of these privacy application-based research papers, people utilize DP on top of FL in order to calibrate noise into the model weights before they are uploaded. Thus, all updates the server receives are "differentially indistinguishable" from one another as a mathematical privacy guarantee. We saw in one of the papers (in Ruzafa-Alcazar et al. (2023)) apply both Gaussian and Laplacian noise mechanisms to these model weights. So, in an FL-enabled IDS context for Industrial IoT, research papers often indicate that privacy protection can be maintained without significantly degrading attack detection accuracy.

Application 3: Privacy-preserving federated learning in healthcare.

DP is undeniably and critically researched here, due to strict regulations like HIPPA and GDPR. Teo et al. (2024) finds that DP in FL models in healthcare are still susceptible to membership inference attacks and reconstruction attacks. In these types of attacks, an adversary can infer whether a specific patient's data was used in training, or even reconstruct back the original patient record entirely. So, naturally, DP and FL models in clinical settings are one of the hardest environments/contexts to apply DP, because the cost of getting the wrong privacy-utility tradeoff is measured in patient outcomes rather than just "performance."

References (Everything Looked at)

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